

1. Model Implementation and Results

In this section, several classification models were implemented to predict biodegradability using the selected molecular descriptors. Both linear and nonlinear methods were considered to evaluate how model complexity affects performance. All models were trained using the same training dataset and evaluated on the same test dataset to ensure fair comparison. Hyperparameter tuning was performed where applicable using cross validation to improve model performance and reduce overfitting.

4.1 Logistic Regression

Model

$$\text{logit}(P(Y = RB | X)) = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p$$

Variables

All selected predictors after removing near-zero variance and highly correlated features were used.

Variable Selection

Feature selection was performed prior to modeling by removing near-zero variance and highly correlated variables, ensuring that only informative predictors were included.

Tuning

Logistic regression was further tuned using L1 regularization (lasso) via cross validation to select the optimal lambda value. Hyperparameter tuning was performed using cross-validation to improve model performance and prevent overfitting.

Interpretation

Logistic regression achieved the highest test accuracy, indicating that the relationship between predictors and biodegradability can be effectively captured using a linear model after preprocessing. The model's coefficients were estimated using the training data. Several predictors showed non-zero coefficients after L1 regularization, indicating their contribution to the classification model. This result suggests that, after preprocessing, the underlying relationship between predictors and biodegradability is relatively linear.

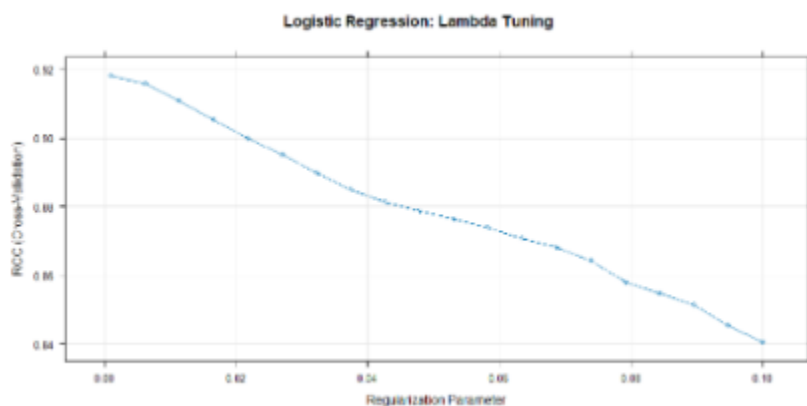


Figure 6: Cross validation results for logistic regression regularization parameter.

4.2 Linear Discriminant Analysis (LDA)

Model

LDA assumes that each class follows a multivariate normal distribution and uses linear decision boundaries.

Variable Selection

The same preprocessed data set was used to ensure consistency across models.

Tuning

Hyperparameter tuning was performed using cross-validation to improve model performance and prevent overfitting. No major tuning parameters were required for LDA.

Interpretation

LDA performed slightly worse than logistic regression, suggesting that its assumptions about class distributions may not fully hold for this dataset. Additionally, the overlap observed in the EDA suggests that the classes may not be well separated by a single linear boundary under the assumptions of LDA.

4.3 Random Forest

Model

Random Forest builds multiple decision trees and aggregates predictions to improve accuracy.

Variable Selection

The model was applied to the same selected feature set after preprocessing.

Tuning

Hyperparameters such as mtry and ntree were tuned using cross validation. This was done to improve model performance and prevent overfitting.

Interpretation

Random forest performed well but did not outperform logistic regression, suggesting that complex nonlinear relationships were not necessary after preprocessing. This indicates that the additional flexibility of Random Forest did not provide significant benefits once the data was cleaned.

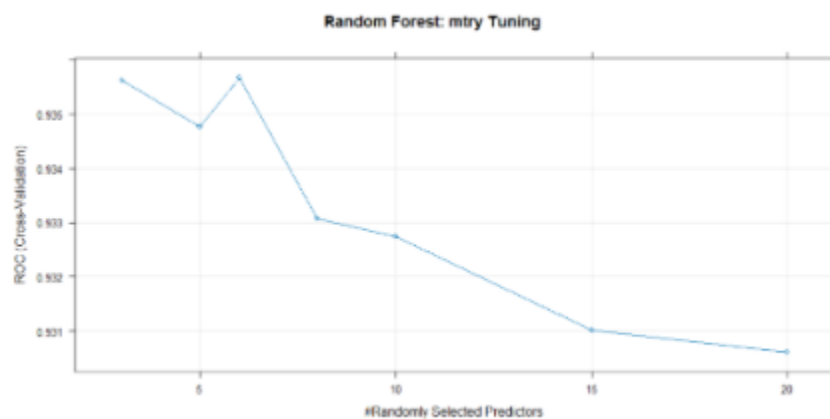


Figure 7: Cross validation results for Random Forest hyperparameter tuning

4.4 Support Vector Machine (SVM)

Model

A radial kernel SVM was used to capture nonlinear decision boundaries.

Variable Selection

The same preprocessed data was used.

Tuning

Hyperparameters including cost (C) and gamma were tuned using cross validation. This was done to improve model performance and prevent overfitting.

Interpretation

SVM performed comparably to Random Forest but did not exceed logistic regression performance, indicating that nonlinear flexibility provided limited additional benefit. This further supports the conclusion that nonlinear complexity was not necessary for this dataset.